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ACRME Placement Scoring Weights — Complete Reference

Project: Azure Capacity Reservation Management Engine (ACRME)

Classification: Technical Reference

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Overview

The ACRME placement engine uses **three environment-type-specific scoring formulas** (PS_Prod, PS_NonProd, PS_DR) to rank Azure regions for Prod, NonProd/CVAL, and DR environment placement. All three formulas share the same **default weight values**:

- = 0.30 (CRG Capacity)
- = 0.20 (Quota Headroom)
- = 0.25 (Distribution Fairness)
- = 0.15 (DR/Overflow Health)
- = 0.10 (Zone Diversity)

However, and have **different semantic meanings** per environment type — they measure different signals depending on whether you’re scoring a Prod, NonProd, or DR region.

Weight Definitions by Component

(Alpha) — CRG Capacity Signal [Weight: 0.30]

Purpose: Measures available capacity in the region's CRG most relevant to the environment being placed.

Semantic per environment:

Environment	Measures	Formula
PS_Prod	NonProd CRG overflow headroom relative to Prod CRG size. High = the associated NonProd CRG has capacity to absorb overflow if Prod grows.	$\text{nonprod_crg_effective_free} / \text{prod_crg_quantity}$
PS_NonProd	Fraction of NonProd CRG that is truly available (free_slots minus dr_overflow_reserve). This is the primary capacity signal for NonProd placement.	$\text{nonprod_crg_effective_free} / \text{nonprod_crg_quantity}$
PS_DR	Fraction of DR CRG capacity that is uncommitted. High = more room for additional DR assignments.	$\text{dr_crg_free_slots} / \text{dr_crg_quantity}$

Why is the highest weight (0.30): Capacity is the most critical constraint. A region with zero headroom cannot accept new placements regardless of quota or distribution. is highest when capacity is tight; can be reduced to 0.20 once regions reach steady-state utilization (40–60%).

(Beta) — Quota Headroom [Weight: 0.20]

Purpose: Measures available Azure subscription/group quota headroom for the relevant CRG.

Semantic per environment:

Environment	Measures	Formula
PS_Prod	Prod group quota headroom ratio	$\text{prod_crg_quota_headroom_vcpu} / \text{prod_crg_quota_limit_vcpu}$
PS_NonProd	NonProd floor-adjusted quota headroom (effective ceiling minus used)	$\text{nonprod_crg_quota_headroom_vcpu} / \text{nonprod_crg_quota_limit_vcpu}$
PS_DR	DR quota headroom within the DR floor allocation	$\text{dr_crg_quota_headroom_vcpu} / \text{dr_crg_quota_limit_vcpu}$

Why = 0.20: Quota exhaustion silently blocks CR creation and VM deployment. ensures the engine directs placements toward regions with quota headroom. Lower than because quota can be requested (takes time but is not a hard ceiling like physical capacity).

Note: All calculations read from **quota-group-level fields** (two-group model: Prod-only group, shared NonProd+DR group), not legacy per-subscription quota. See ADR-002.

(Gamma) — Distribution Fairness [Weight: 0.25]

Purpose: Acts as a **self-correcting load balancer**. Rewards regions with fewer customers already placed, driving even distribution across regions over time.

Semantic (same for all environments):

$$= 1 - (\text{customer_count_in_this_region} / \text{total_customers})$$

Environment	Measures	Formula
PS_Prod	Prod region fairness	$1 - (\text{consumer_count_prod} / \text{total_customers})$
PS_NonProd	NonProd region fairness	$1 - (\text{consumer_count_nonprod} / \text{total_customers})$
PS_DR	DR region fairness	$1 - (\text{consumer_count_dr} / \text{total_customers})$

Why $\gamma = 0.25$: Equal to capacity in importance. As a region accumulates customers its score drops, redirecting future placements away from it until all regions reach roughly equal count (steady-state expected count per region per env type = $\text{total_customers} / 3$). Increase toward 0.35 if score convergence results in uneven loading; decrease toward 0.15 if capacity constraints should dominate fairness.

Note: The PRR (§28) updates Distribution to use **demand units** rather than customer count: $1 - (\text{Region_Assigned_Demand} / \text{Total_Assigned_Demand})$.

(Delta) — DR/Overflow Health [Weight: 0.15]

Purpose: Environment-specific **resilience signal** — measures DR coverage readiness (Prod/DR) or overflow capacity health (NonProd).

Semantic per environment:

Environment	Measures	Formula	Intent
PS_Prod	DR CRG coverage readiness of this region's own DR CRG. When scoring a Prod region, captures the health of the region's DR CRG — a Prod region whose DR CRG is well-covered scores higher.	$\text{dr_crg_coverage_ratio}$	Prod placement favors regions with healthy DR capacity

Environment	Measures	Formula	Intent
PS_NonProd	Overflow capacity health — intentionally shares the same ratio as . Rewards regions genuinely under-utilised from both a NonProd-placement and a DR-overflow perspective.	$\text{nonprod_crg_effectiveness} / \text{nonprod_crg_quantity}$ (same as)	NonProd placement prioritizes regions with DR overflow headroom
PS_DR	Coverage ratio health relative to target. Scores 1.0 when coverage dr_ratio_max (0.40). Proportionally below 1.0 for coverage between dr_ratio_min (0.30) and dr_ratio_max (e.g. $0.35/0.40 = 0.875$).	$\min(1.0, \text{dr_crg_coverage_ratio} / \text{dr_ratio_max})$	DR placement drives buffer margins toward target

Why is different per env: Each environment has a distinct resilience need. Prod cares about its paired DR region’s readiness; NonProd cares about overflow headroom it can yield to DR during a crisis; DR cares about its own coverage health.

Why = 0.15: Lower than / / because it’s a forward-looking health signal rather than an immediate placement blocker. Increase toward 0.20 if DR buffer violations appear; decrease if capacity/distribution should dominate.

Trade-off (PS_NonProd): duplicates intentionally — this was flagged in the PRR (§28) as double-counting. The corrected pilot formula proposes revised NonProd weights: $\text{PS_NonProd} = 0.35 \cdot \text{Capacity} + 0.25 \cdot \text{Quota} + 0.25 \cdot \text{Distribution} + 0.05 \cdot \text{DR_Overflow_Integrity} + 0.10 \cdot \text{Zones}$ to reduce the duplication.

(Epsilon) — Zone Diversity [Weight: 0.10]

Purpose: Rewards regions with more availability zones, ensuring placed workloads can leverage Azure’s zone-redundancy features.

Semantic (same for all environments):

$$= \text{az_count} / 3$$

Normalizes to 1.0 for a 3-zone region (maximum diversity); proportionally lower for 2-zone (0.667) or 1-zone (0.333) regions.

Environment	Measures	Formula
All (PS_Prod, PS_NonProd, PS_DR)	Zone diversity	$\text{az_count} / 3$

Why = 0.10 (lowest weight): Zone diversity is desirable but not a placement blocker. Hard constraint **HC-5 ZONE_AVAILABILITY** already enforces a minimum zone count — acts as a tiebreaker that favors 3-zone regions when all else is equal. Increase toward 0.15 if zone diversity is a critical operational requirement; keep at 0.10 if capacity/distribution dominate.

Complete Formula Definitions

PS_Prod(r) — Production Region Score

PS_Prod(r) =

0.30 × (nonprod_crg_effective_free / prod_crg_quantity)	← : NonProd overflow headroom
+ 0.20 × (prod_crg_quota_headroom_vcpu / prod_crg_quota_limit_vcpu)	← : Prod quota headroom
+ 0.25 × (1 - consumer_count_prod / total_customers)	← : Prod distribution fairness
+ 0.15 × dr_crg_coverage_ratio	← : DR readiness of likely CRG
+ 0.10 × (az_count / 3)	← : zone diversity

Guards: - If prod_crg_quantity == 0 → PS_Prod = 0.0 - If prod_crg_quota_limit_vcpu == 0 → term = 0.0 - If dr_crg_coverage_ratio is None → term = 0.0 (no DR CRG established yet)

PS_NonProd(r) — Non-Production / CVAL Region Score

Current formula (with intentional / duplication):

PS_NonProd(r) =

0.30 × (nonprod_crg_effective_free / nonprod_crg_quantity)	← : effective NonProd capacity
+ 0.20 × (nonprod_crg_quota_headroom_vcpu / nonprod_crg_quota_limit_vcpu)	← : NonProd quota headroom
+ 0.25 × (1 - consumer_count_nonprod / total_customers)	← : NonProd distribution fairness
+ 0.15 × (nonprod_crg_effective_free / nonprod_crg_quantity)	← : overflow capacity
+ 0.10 × (az_count / 3)	← : zone diversity

Proposed pilot formula (corrected to avoid double-counting):

PS_NonProd(r) =

0.35 × (nonprod_crg_effective_free / nonprod_crg_quantity)	← : capacity (increased)
+ 0.25 × (nonprod_crg_quota_headroom_vcpu / nonprod_crg_quota_limit_vcpu)	← : quota headroom
+ 0.25 × (1 - consumer_count_nonprod / total_customers)	← : distribution
+ 0.05 × (nonprod_crg_effective_free / nonprod_crg_quantity)	← : DR overflow integ.
+ 0.10 × (az_count / 3)	← : zone diversity

Guards: - If nonprod_crg_quantity == 0 → PS_NonProd = 0.0 - If nonprod_crg_quota_limit_vcpu == 0 → term = 0.0

PS_DR(r) — Disaster Recovery Region Score

PS_DR(r) =

0.30 × (dr_crg_free_slots / dr_crg_quantity)	← : DR CRG headroom
+ 0.20 × (dr_crg_quota_headroom_vcpu / dr_crg_quota_limit_vcpu)	← : DR quota headroom
+ 0.25 × (1 - consumer_count_dr / total_customers)	← : DR distribution
+ 0.15 × min(1.0, dr_crg_coverage_ratio / dr_ratio_max)	← : coverage ratio headroom

$$+ 0.10 \times (\text{az_count} / 3)$$

← : zone diversity

Guards: - If `dr_crg_quantity == 0` → `PS_DR = 0.0` (no DR CRG — region ineligible) - If `dr_crg_quota_limit_vcpu == 0` → `term = 0.0` - If `dr_crg_coverage_ratio` is `None` → `term = 0.0`

Weight Summary Table

Component	Symbol	Weight	PS_Prod Measures	PS_NonProd Measures	PS_DR Measures
CRG		0.30	NonProd overflow	NonProd effective	DR CRG free
Capacity			headroom	headroom	capacity
Quota		0.20	Prod group quota	NonProd floor-adjusted	DR quota within
Headroom			ratio	quota	floor
Distribution		0.25	Prod region fairness	NonProd region	DR region
				fairness	fairness
DR/Overflow		0.15	DR coverage	Overflow capacity	Coverage vs
Health			readiness	(same as)	target ratio
Zone		0.10	<code>az_count / 3</code>	<code>az_count / 3</code>	<code>az_count / 3</code>
Diversity					

Clamping & Score Range

Every raw component ratio is **clamped** before weighting:

$$\text{Clamp}(x) = \max(0, \min(1, x))$$

This ensures each component contributes a value in the range $[0.0, 1.0]$, and the final weighted score for any region is:

$$\text{PS}(r) \in [0.0, 1.0]$$

A uniform jitter of ± 0.01 is added as a tiebreaker.

Tuning Guidance

When to adjust (Capacity)

- **Increase** toward **0.35–0.40** when capacity is tight and regions are unevenly provisioned.
- **Reduce** toward **0.20** once regions reach steady-state utilization (40–60%) and distribution becomes the primary concern.

When to adjust (Distribution)

- **Increase** toward **0.35** if score convergence results in uneven region loading (some regions accumulate too many customers).
- **Decrease** toward **0.15** if capacity constraints should dominate fairness (tight capacity environments).

When to adjust (DR/Overflow Health)

- **Increase toward 0.20** if DR buffer violations or coverage gaps appear frequently.
- **Reduce toward 0.10** if capacity/distribution should dominate and DR health is consistently good.

When to adjust (Zone Diversity)

- **Increase toward 0.15** if zone diversity is a critical operational requirement (e.g., all placements must strongly favor 3-zone regions).
- **Keep at 0.10** (default) if capacity, quota, and distribution are the dominant concerns.

Default recommendation: Start with the default weights ($=0.30$, $=0.20$, $=0.25$, $=0.15$, $=0.10$) for the pilot. After 30–50 customer placements, analyze the score breakdown data in `OperationRecord` and adjust incrementally based on observed convergence behavior.

Source References

- **Multi-Region Placement Design** (docs/research/multi_region_placement_design.md) — lines 449–531, full formula definitions + weight table
 - **Design Change Summary** (docs/research/design_change_summary.md) — CR/CRG-7, lines 153–193, formula rationale
 - **Production Readiness Review** (docs/acrme_production_readiness_review_and_architecture.md) — §28 lines 1095–1160, corrected scoring model
 - **ADR-001 Region Selection** (docs/adr/acrme_adr_001_region_selection.md) — corrected scoring model section
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Evidence Tags

- [Documented] — Defined in Multi-Region Placement Design and PRR §28
 - [Decided] — Decision Log D9 (env-type-specific formulas)
 - [Assumed] — Default weights retained for pilot comparison; revised weights proposed but not empirically validated
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Document Status: Reference

Next Review: After pilot Phase 4 (policy tuning with 30–50 placements measured)